

Mamdouh Attia

Mobile Software Engineer [Flutter] | Fresh Graduate

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EDUCATION

Cairo University

Bachelor of Engineering, Computer Engineering Department

- Very Good with Honors.

Cairo, Egypt

2019 - 2024

EXPERIENCE

Machine Learning Engineer Instructor

DEPI Initiative, Ministry of Communications and Information Technology

- Delivered hands-on machine learning training sessions for aspiring engineers.
- Designed curriculum and practical exercises to cover essential ML concepts and tools.

Aug. 2024 – Oct. 2024

Hybrid

Flutter Developer (Freelance)

Self-employed

- Developed mobile applications for 5+ clients using Flutter.
- Collaborated with clients to gather requirements and deliver tailored solutions.

Aug. 2023 – Present

Hybrid

Flutter Instructor

Primalab

- Taught Dart programming and Flutter essentials to beginners, including layout, widgets, navigation, and routing.
- Guided students through project-based learning and requirements gathering.

Aug. 2023 – Feb. 2024

New Damietta, Egypt

Flutter Developer Intern

Orange Digital Center Software House | [Certificate Link](#)

- Worked with agile teams on two mobile app projects.
- Translated Figma designs into Flutter UI, meeting tight deadlines.
- Developed apps using MVVM architecture and Bloc Cubit.

Jun. 2023 – Sep. 2023

Remote

FLUTTER PROJECTS

Reddit Clone | Flutter, MVC-S, Agile, REST APIs | [Repo Link](#) | [Demo Link](#)

Aug. 2022 – Dec. 2022

- Developed a Reddit mobile app clone as part of a 6-member team in a software house environment.
- Implemented UI components including drawers, settings, and community creation screens.
- Contributed to unit testing, mock services, documentation, and backend integration.
- Followed Agile methodology with dedicated roles for frontend, backend, DevOps, and QA.

Check-It | Flutter, Barcode, E-commerce, Government Integration

Feb. 2024 – Jun. 2024

- Developed a mobile solution to verify product prices and combat price manipulation in retail.
- Implemented barcode scanning, real-time price comparison, and e-commerce style product database.
- Enabled users to submit complaints and track their status with government authorities.

VolunTeen | Flutter, Firebase, GPT-3, Responsive Design

Sep. 2023 – Jan. 2024

- Solo project for a real client: built a volunteering app targeting teenagers, featuring 30–40 screens.
- Integrated AI chatbot (GPT-3), in-app video player, and third-party services.
- Implemented MVC architecture, Firebase (auth, storage), shared preferences, and WhatsApp/URL launcher.
- Designed and delivered all UI/UX, content, and backend integrations.

HyperMate | Flutter, ESP8266, GSR Sensor, AI Chatbot

Nov. 2023 – Dec. 2023

- Solo client project: developed a stress monitoring app connecting to ESP8266 microcontroller via Wi-Fi.
- Displayed real-time GSR sensor readings and provided personalized exercise and nutrition suggestions.
- Integrated a smart chatbot for user support and recommendations.

EcoSort | Flutter, Firebase, Blync API, QR, Notifications

Apr. 2024 – May. 2024

- Enhanced an existing client app by adding Firebase authentication, push notifications, and Blync API integration.
- Implemented QR code scanning and a point reward system for recycling activities.

GRADUATION PROJECT

- Predicto** | *Python, OpenCV, Machine Learning* | [Demo video](#) Aug. 2023 – July 2024
- Developed an open-source Python library named 'vPredicto', specialized in video frame prediction.
 - Integrated the latest state-of-the-art Deep Learning models in this field.

OTHER PROJECTS

- Web-page Phishing Detection** | *Python, Machine Learning* | [Repo Link](#) Apr. 2024 – May. 2024
- Developed a phishing detection system by training and evaluating 12 machine learning models with hyperparameter tuning.
 - Applied advanced evaluation metrics and ML concepts to address real-world cybersecurity threats.
- Gaming Engine + Drive Me Home** | *C++, OpenGL, OOP* | [Repo Link](#) | [Demo Link](#) Mar. 2023 – May. 2023
- Co-developed a custom game engine & a game using C++ & OpenGL, implementing shaders, textures & lighting.
 - Engine: Designed shader, texture, sampler, and material classes.
 - Game: Built collision system, post-processing shaders, menus, level-up/game-over states & multiple game objects.
- RSA E2E Encrypted Chat** | *Python, Security, Socket Programming* | [Repo Link](#) | [Explanation video](#) Apr. 2023
- Created a secure Python chat service with end-to-end RSA encryption using sockets.
- Logic Simulator** | *C++, OOP, CMU Graphics* | [Repo Link](#) Apr. 2021
- Team project: Developed a Windows logic simulator for circuit design and simulation with features like save, copy, undo/redo.
- Rovers Mars-station** | *C++, Data Structures & Algorithms* | [Repo Link](#) Dec. 2021
- Team project: Implemented mission scheduling and operations for Mars exploration using advanced data structures and algorithms.
- Search Engine** | *Java, OOP, HTML/CSS/JS* | [Repo Link](#) Apr. 2022
- Team project: Built a Java-based search engine with a web interface.
 - Focused on implementing the search engine crawler in Java.
- Covid Detector** | *Python, Deep Learning, TensorFlow, Keras* | [Repo Link](#) Sep. 2021
- Built a deep learning model to detect Covid-19 from X-ray images using VGG16, VGG19, and AlexNet architectures.
 - Applied advanced DL concepts and evaluated model performance on real datasets.
- MIPS Processor** | *Verilog, Computer Architecture* | [Repo Link](#) Dec. 2022
- Team project: Implemented a pipelined MIPS processor in Verilog.
 - Responsible for the Decode stage and system integration.
- OS Scheduler** | *C, Operating Systems* Apr. 2022
- Team project: Developed an OS-like scheduler implementing process scheduling algorithms in C on Linux.

SKILLS

- Programming:** C/C++ | Dart | Python | Java
- Mobile:** Flutter | Provider | Bloc/Cubit | MVC | MVVM | Clean Architecture | Hive | Shared Prefs | Firebase | Supabase | API Integration | Unit Testing | Documentation
- AI:** ML/DL | Computer Vision | NLP | Keras | TensorFlow | PyTorch | Azure AI
- Other:** SOLID | OOP | Data Structures | Algorithms | Design Patterns | Clean Code | Linux | Git/GitHub

CERTIFICATES

- Coursera's ML Specialization** | [Certificate Link](#) 2024
- Huawei Artificial Intelligence HCIA-AI** 2023

VOLUNTEERING

Instructor to the Staff of My City's Orphanage

Damietta, Egypt

- Windows Fundamentals Instructor Aug. 2021 – Sep. 2021
- Microsoft Word Fundamentals Instructor Aug. 2022 – Sep. 2022